

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Domperidone and metoclopramide both block D2 receptors — why is domperidone preferred in Parkinson's disease?
2. Which antiemetic class is best for the delayed phase of chemo-induced vomiting, and which enzyme governs its main drug interactions?
3. Name the two emetic drugs on this lecture and one contraindication for each.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The reflex & the transmitters behind it

Vomiting (emesis) is the forceful expulsion of gastric contents, usually preceded by nausea. Causes: gastritis, peptic ulcer, intestinal obstruction, drug-induced, pregnancy, severe pain, brain tumour, iatrogenic, and more. Four neurotransmitter/receptor pairs drive the reflex, and every antiemetic class on this lecture is named for the one it blocks.

Transmitter	Receptor
Serotonin (5-HT)	5-HT ₃
Acetylcholine	Muscarinic (M)
Dopamine	D ₂
Histamine	H ₁

2 Emetic drugs (used to induce, not treat, vomiting)

Apomorphine	Semi-synthetic morphine derivative. IM/SC, acts centrally. Dose 6 mg (2–8 mg). Induces vomiting in 5–10 minutes. Contraindicated in respiratory failure (CNS depressant effect).
Ipecacuanha	Contains emetine and cephaline. Given as syrup; vomiting in ~15 minutes. Acts by irritating the gastric mucosa and the CTZ. Dose: 5 mL infants, 10–15 mL children, 20 mL adults. Contraindicated in kerosene/corrosive poisoning, unconscious patients, morphine poisoning.

3 Antiemetic drugs — receptor antagonists

5-HT₃ antagonists (granisetron, ondansetron, dolasetron, palonosetron)	Primary site: CTZ. Particularly valuable for postoperative, radiation-, and cytotoxic-induced (e.g. cisplatin) vomiting — more effective on vomiting than on nausea itself. Oral or injection. Excreted by kidney and liver — no renal dose reduction needed, but reduce in liver impairment. Tissue half-life 4–9h (palonosetron 40h). ADR: headache, GI upsets (especially constipation), QT-interval prolongation (more with dolasetron).
Metoclopramide (substituted benzamide)	D ₂ antagonist, closely related to phenothiazines. Acts centrally on the CTZ and peripherally on the GI tract — increases motility of oesophagus, stomach, and intestine, which also makes it useful in GERD and hepatic/biliary disorders. Dose 10–20 mg every 6h, oral or parenteral. Rapid, complete oral absorption; high first-pass metabolism; excreted in urine (free + metabolites) and in breast milk. ADR: movement disorders (more common in children/ young adults) — motor restlessness, spasmodic torticollis, oculogyric crises; also stimulates prolactin → galactorrhoea, menstrual disorders, impotence, gynaecomastia.
Domperidone	Similar use to metoclopramide (cytotoxic-induced vomiting, GI symptoms) but does not readily cross the blood-brain barrier — far less prone to central/extrapyramidal side effects, which is exactly why it's preferred in Parkinson's disease. Carries a small increased risk of serious cardiac adverse effects (especially higher doses, older patients) — use now restricted. Oral, half-life 4–5h, excreted in urine.
Phenothiazines (chlorpromazine, perphenazine, prochlorperazine, trifluoperazine)	Antipsychotics that are also effective antiemetics — used for more severe N&V from cancer, radiation, cytotoxics, opioids, anaesthetics. Oral, IV, or suppository. Mainly D ₂ antagonists in the CTZ, but also block histamine and muscarinic receptors. ADR: sedation (especially chlorpromazine), hypotension, extrapyramidal symptoms (dystonias, tardive dyskinesia). Haloperidol, droperidol, and levomepromazine act the same way and are used for acute CINV.
NK₁ antagonists (aprepitant, rolapitant, fosaprepitant, netupitant)	Block substance-P (NK ₁) receptors in the CTZ and vomiting centre. Aprepitant is oral and controls the delayed phase of cytotoxic-drug emesis specifically. Aprepitant/rolapitant are metabolised by CYP3A4 — avoid strong CYP3A4 inhibitors (e.g. clarithromycin) or inducers (e.g. St John's wort). Aprepitant itself induces CYP3A4/2C9 and dose-dependently inhibits CYP3A4 — numerous interactions. Rolapitant is a moderate CYP2D6 inhibitor. Fosaprepitant is the IV prodrug of aprepitant. ADR: fatigue, diarrhoea, abdominal pain, hiccups.

The rest of the antiemetic toolkit

Class	Drugs	Key points
H ₁ antagonists	Cinnarizine, cyclizine, promethazine	Effective against many causes incl. motion sickness and gastric irritants — not effective against CTZ-direct causes. Promethazine: severe pregnancy morning sickness, NASA's choice for space motion sickness. ADR: drowsiness/sedation.
Muscarinic antagonists	Hyoscine (scopolamine)	Prophylaxis/treatment of motion sickness — oral or transdermal patch. ADR: dry mouth, blurred vision; less sedating than antihistamines (poor CNS penetration).
Cannabinoids	Nabilone	Reduces CTZ-driven vomiting, sometimes effective where others fail. Oral, well absorbed, half-life ~120 min. ADR: drowsiness, dizziness, dry mouth, mood changes, postural hypotension. Antiemetic effect is antagonised by naloxone — implies opioid receptors play a role.
Glucocorticoids	Dexamethasone (high-dose)	Controls emesis, especially cytotoxic-induced. Mechanism unclear — possibly prostaglandin blockade. Typically combined with metoclopramide or ondansetron.
Benzodiazepines	Alprazolam, lorazepam	Used as adjuncts in combination regimens.

4 Combination regimens

Corticosteroids (usually dexamethasone) boost antiemetic activity when combined with high-dose metoclopramide, a 5-HT₃ antagonist, a phenothiazine, a butyrophenone, or a benzodiazepine. Antihistamines (e.g. diphenhydramine) are added to high-dose metoclopramide to reduce extrapyramidal reactions, or to corticosteroids to counter metoclopramide-induced diarrhoea. Adding an NK₁ antagonist to a 5-HT₃ antagonist + dexamethasone is especially useful for highly emetogenic regimens, particularly **delayed CINV**.

Combination (first 24h post-cisplatin)	Response rate
Dexamethasone + Ondansetron	91%
Dexamethasone + Diphenhydramine + Droperidol + Metoclopramide	76%
Dexamethasone + Lorazepam + Metoclopramide	63%
Dexamethasone + Diphenhydramine + Metoclopramide	58%

5 Clinical uses, quick reference

Drug	Used for
Cyclizine	Motion sickness, vestibular disorders, surgery/postoperative narcotic-induced N&V
Cinnarizine	Motion sickness, vestibular disorders (e.g. Ménière's disease)
Promethazine	Severe morning sickness of pregnancy, motion sickness, vestibular disorders
Hyoscine	Motion sickness
Prochlorperazine (phenothiazine)	Migraine, vestibular disorders, radiation, viral gastroenteritis, severe morning sickness
Metoclopramide	Migraine, radiation, GI disorders, cytotoxic drugs, prevention of postoperative N&V
Domperidone	Less CNS side effects in Parkinson's patients — poor BBB penetration
Ondansetron	Cytotoxic drugs or radiation, postoperative vomiting
Nabilone	Cytotoxic drugs
Fosaprepitant	Cytotoxic drugs (delayed CINV)

Mnemonic — receptor-to-class: “Her Mum Drove 5 Cars, Nowhere.” H₁ / M / D₂ / 5-HT₃ / CB₁ / NK₁.

P Must-remember pearls

- Domperidone barely crosses the BBB — that's exactly why it's the D₂ antagonist of choice in Parkinson's disease.
- Metoclopramide's signature side effects are extrapyramidal (movement disorders) and hyperprolactinaemia — both come from blocking D₂ elsewhere in the CNS.
- 5-HT₃ antagonists are dose-reduced for liver impairment, not renal impairment.
- Aprepitant is the one to remember for delayed CINV, and for its CYP3A4-heavy interaction profile.
- Nabilone's antiemetic effect is reversed by naloxone — a clue that opioid receptors are involved in its mechanism.

Q Test yourself

1. A Parkinson's disease patient needs an antiemetic. Why is domperidone favoured over metoclopramide?

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|--|---|
| A) Domperidone is a 5-HT ₃ antagonist | B) Domperidone poorly crosses the blood-brain barrier, avoiding central dopamine blockade |
| C) Metoclopramide is not a D ₂ antagonist | D) Domperidone has no cardiac risk at all |

2. A chemotherapy patient needs coverage for delayed-phase CINV specifically. Which class is best suited?

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|-------------------------------|--|
| A) H ₁ antagonists | B) NK ₁ antagonists (e.g. aprepitant) |
| C) Muscarinic antagonists | D) Emetics |

3. A young patient on metoclopramide develops involuntary upward eye movements. What is this called, and why does it happen?

- A) Oculogyric crisis, from central D2 blockade
- B) Tardive dyskinesia, from H1 blockade
- C) Serotonin syndrome, from 5-HT₃ blockade
- D) A normal expected effect, no action needed

4. A patient is given nabilone for refractory chemo-induced nausea, then separately given naloxone for an unrelated reason. What happens to the antiemetic effect?

- A) It is unaffected
- B) It is antagonised — naloxone blocks the opioid receptor component of nabilone's action
- C) It is potentiated
- D) Naloxone converts it to an emetic

5. A patient with unintentional kerosene ingestion is about to be given ipecacuanha to induce vomiting. Why should this be avoided?

- A) Ipecacuanha is contraindicated in corrosive/kerosene poisoning
- B) Ipecacuanha only works in adults
- C) Ipecacuanha has no CTZ action
- D) Ipecacuanha requires IV administration

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B — domperidone's poor blood-brain barrier penetration spares central dopamine pathways, avoiding the extrapyramidal worsening metoclopramide can cause.

2 — B — aprepitant (an NK1 antagonist) is specifically noted for controlling the late/delayed phase of cytotoxic-drug emesis.

3 — A — an oculogyric crisis, one of the extrapyramidal reactions caused by metoclopramide's central D2 blockade.

4 — B — naloxone antagonises nabilone's antiemetic effect, implying opioid receptors contribute to its mechanism.

5 — A — ipecacuanha is contraindicated in kerosene/corrosive poisoning (as well as in unconscious patients and morphine poisoning).

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Jangi Shawkat Salai lecture slides, LG 1 (full deck).
Study alongside the lecture, not instead of it.